

5-1 · Determine the Area of Parallelograms and Rhombuses

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute 24×13 .

312

$$24 \times 10 + 24 \times 3 = 240 + 72.$$

Skill: Multiply two- and three-digit numbers

2. FOUNDATION 2

A rectangle is 8 cm long and 6 cm wide. Find its area in square centimeters.

48

Multiply length by width: $8 \times 6 = 48$.

Skill: Find the area of a rectangle by multiplying length by width

3. FOUNDATION 3

A parallelogram has base 10 cm, slanted side 7 cm, and perpendicular height 4 cm. Which number is the height?

4

Height meets the base at a right angle; the slanted side is not the height.

Skill: Tell a base from a slanted side and identify the perpendicular height

4. FOUNDATION 4

A rectangle measures 3 meters by 5 meters. State its area with units.

15 square meters

$3 \times 5 = 15$; area counts square units.

Skill: Label an area answer in square units

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5. CHECK A

Area of an 8 cm by 5 cm rectangle? In a parallelogram, is the height the slanted side?

**Rectangle area = $8 \text{ cm} \times 5 \text{ cm} = 40 \text{ cm}^2$.
Height question: No! The height must be perpendicular to the base, never the slanted side.**

Compare each part of your work with the worked solution.

6. CHECK B

Area of a 9 cm by 7 cm rectangle? In a parallelogram, is the height the slanted side?

$9 \times 7 = 63 \text{ cm}^2$. No — the height must meet the base at a right angle; the slanted side is longer than the height.

Compare each part of your work with the worked solution.

7. CHECK C

Area of a 12 m by 4 m rectangle? A parallelogram has base 12 m, slant side 5 m, height 4 m. What is its area?

Rectangle: $12 \times 4 = 48 \text{ m}^2$. Parallelogram: $12 \times 4 = 48 \text{ m}^2$ — use the height 4, not the slant 5.

Compare each part of your work with the worked solution.

8. TRANSFER

Two parallelograms both have base 10 cm and height 6 cm but different slant sides. Compare their areas.

Both are 60 cm^2 . The slant side does not affect area — only base and perpendicular height do.

Explain why your method works and include units when needed.

5-2 · Determine the Area of Triangles

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $(14 \times 9) \div 2$.

63

Halve 14 first to get 7, then multiply 7×9 .

Skill: Multiply, then halve — or halve, then multiply

2. FOUNDATION 2

Compute $\frac{1}{2} \times 38$.

19

Multiplying by one half is the same as dividing by 2.

Skill: Multiply a whole number by $\frac{1}{2}$

3. FOUNDATION 3

A parallelogram has base 12 cm and height 5 cm.
Find its area in square centimeters.

60

Area is base $\times$ perpendicular height: 12×5 .

Skill: Find the area of a rectangle or parallelogram

4. FOUNDATION 4

A triangle has base 8 cm, perpendicular height 6 cm, and slanted side 10 cm. Which length belongs in the height part of its area formula?

6

Use the length perpendicular to the chosen base.

Skill: Identify the height drawn perpendicular to the chosen base

5-2 · Determine the Area of Triangles

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

7×4 , then half? Area of a triangle with base 7 m and height 4 m?

$7 \times 4 = 28$, then half = 14. Triangle area: $A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 7 \text{ m} \times 4 \text{ m} = 14 \text{ m}^2$.

Compare each part of your work with the worked solution.

6. CHECK B

9×6 , then half? Area of a triangle with base 9 cm and height 6 cm?

$9 \times 6 = 54$, half is 27. Area = $\frac{1}{2} \times 9 \times 6 = 27 \text{ cm}^2$.

Compare each part of your work with the worked solution.

7. CHECK C

12×5 , then half? Area of a triangle with base 12 in and height 5 in?

$12 \times 5 = 60$, half is 30. Area = $\frac{1}{2} \times 12 \times 5 = 30 \text{ in}^2$.

Compare each part of your work with the worked solution.

8. TRANSFER

A triangle has area 36 cm^2 and base 12 cm. What is its height?

$36 = \frac{1}{2} \times 12 \times h$, so $36 = 6h$ and $h = 6 \text{ cm}$.

Explain why your method works and include units when needed.

5-3 · Determine the Area of Trapezoids

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $(7 + 11) \times 4$.

72

Add inside parentheses first: $18 \times 4 = 72$.

Skill: Add the two numbers inside parentheses before multiplying

2. FOUNDATION 2

Compute $1/2 \times 54$.

27

Half of 54 is 27.

Skill: Multiply by $1/2$

3. FOUNDATION 3

Evaluate $1/2 \times (6 + 10) \times 5$.

40

First $6 + 10 = 16$, then half of 16 is 8, then $8 \times 5 = 40$.

Skill: Follow order of operations inside a formula

4. FOUNDATION 4

A shape is made of a rectangle of area 24 and two triangles of area 6 each, without overlap. Find the total area.

36

Add the three nonoverlapping areas: $24 + 6 + 6$.

Skill: Split a figure into a rectangle plus triangles

5-3 · Determine the Area of Trapezoids

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$\frac{1}{2} \times (6 + 10) \times 4 = ?$ Do the parentheses first.

Parentheses first: $(6 + 10) = 16$. Area = $\frac{1}{2} \times 16 \times 4 = 8 \times 4 = 32$ square units.

Compare each part of your work with the worked solution.

6. CHECK B

$\frac{1}{2} \times (8 + 12) \times 5 = ?$ Do the parentheses first.

$(8 + 12) = 20$. Then $\frac{1}{2} \times 20 \times 5 = 10 \times 5 = 50$ square units.

Compare each part of your work with the worked solution.

7. CHECK C

$\frac{1}{2} \times (7 + 9) \times 6 = ?$ Do the parentheses first.

$(7 + 9) = 16$. Then $\frac{1}{2} \times 16 \times 6 = 8 \times 6 = 48$ square units.

Compare each part of your work with the worked solution.

8. TRANSFER

A trapezoid has bases 6 and 10 and area 48. What is its height?

$48 = \frac{1}{2}(6 + 10)h = 8h$, so $h = 6$ units.

Explain why your method works and include units when needed.

5-4 · Apply Area Concepts to Solve Problems

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Find the area of a triangle with base 10 cm and height 8 cm, in square centimeters.

40

Half of 10×8 is 40.

Skill: Find the area of rectangles, triangles, and trapezoids on demand

2. FOUNDATION 2

A rectangle has area 72 square meters. A 15-square-meter section is removed. What area remains?

57

Subtract the removed area: $72 - 15 = 57$.

Skill: Add and subtract areas

3. FOUNDATION 3

A total side length is 14 cm. One part is 9 cm. How long is the other part?

5

Subtract known part from total: $14 - 9 = 5$.

Skill: Find a missing side length from the other labeled sides

4. FOUNDATION 4

Six identical tiles each cover 12 square centimeters. What is the total area?

72

Multiply area per tile by number of tiles: 12×6 .

Skill: Multiply one part's area by the number of identical parts

5-4 · Apply Area Concepts to Solve Problems

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

A 10 by 6 rectangle has a 3 by 2 rectangle cut out. What area is left?

Large rectangle area: $10 \times 6 = 60$. **Cutout rectangle area:** $3 \times 2 = 6$. **Area remaining:** $60 - 6 = 54$ square units.

Compare each part of your work with the worked solution.

6. CHECK B

A 12 by 8 rectangle has a 4 by 3 rectangle cut out. What area is left?

$12 \times 8 = 96$. Cut-out: $4 \times 3 = 12$. **Remaining:** $96 - 12 = 84$ square units.

Compare each part of your work with the worked solution.

7. CHECK C

A 9 by 9 square has a triangle of base 6 and height 4 cut out. What area is left?

Square: 81. **Triangle:** $\frac{1}{2} \times 6 \times 4 = 12$. **Remaining:** $81 - 12 = 69$ square units.

Compare each part of your work with the worked solution.

8. TRANSFER

A rectangular garden 15 m by 10 m has a 3 m by 3 m square shed inside. How much garden is left to plant?

$150 - 9 = 141$ m².

Explain why your method works and include units when needed.

5-5 · Determine the Volume of Rectangular Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.2

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $4 \times 7 \times 5$.

140

Group $4 \times 5 = 20$, then $20 \times 7 = 140$.

Skill: Multiply three factors in any order

2. FOUNDATION 2

Compute $1 \frac{1}{2} \times \frac{2}{3}$.

1

Convert to $\frac{3}{2}$, then multiply $\frac{3}{2} \times \frac{2}{3} = \frac{6}{6}$.

Skill: Multiply fractions and mixed numbers

3. FOUNDATION 3

A rectangular base is 9 cm by 4 cm. Find its area in square centimeters.

36

Multiply $9 \times 4 = 36$ square centimeters.

Skill: Find the area of a rectangular base

4. FOUNDATION 4

A prism measures 2 cm by 3 cm by 5 cm. State its volume with units.

30 cubic centimeters

Multiply $2 \times 3 \times 5 = 30$; volume counts cubic units.

Skill: Label a volume answer in cubic units

5-5 · Determine the Volume of Rectangular Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.2

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

$4 \times 3 \times 2$? $\frac{1}{2} \times \frac{1}{3}$? Volume of a 4 by 3 by 2 prism?

$4 \times 3 \times 2 = 24$. $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. Volume = $4 \times 3 \times 2 = 24$ cubic units.

Compare each part of your work with the worked solution.

6. CHECK B

$5 \times 4 \times 3$? $\frac{1}{2} \times \frac{1}{4}$? Volume of a 5 by 4 by 3 prism?

$5 \times 4 \times 3 = 60$. $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$. Volume = 60 cubic units.

Compare each part of your work with the worked solution.

7. CHECK C

$6 \times 2 \times 4$? $\frac{2}{3} \times \frac{1}{2}$? Volume of a 6 by 2 by 4 prism?

$6 \times 2 \times 4 = 48$. $\frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$. Volume = 48 cubic units.

Compare each part of your work with the worked solution.

8. TRANSFER

A box is $3\frac{1}{2}$ in by 2 in by 4 in. What is its volume?

$3.5 \times 2 = 7$, and $7 \times 4 = 28$ in³.

Explain why your method works and include units when needed.

5-6 · Represent Three-Dimensional Figures in Two Dimensions

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

How many faces, edges, and vertices does a rectangular prism have?

6 faces, 12 edges, 8 vertices

Count the six flat surfaces, twelve joining segments, and eight corners.

Skill: Name the faces, edges, and vertices of a prism and a pyramid

2. FOUNDATION 2

How many faces does a square pyramid have, including its base?

5

One square base plus four triangular faces gives five faces.

Skill: Count the faces of a solid

3. FOUNDATION 3

A net has a rectangular face 3 cm by 7 cm. Find that face's area in square centimeters.

21

Multiply 3×7 ; count this face once.

Skill: Find the area of each rectangle and triangle inside a net

4. FOUNDATION 4

A prism has face pairs sized 3 by 4, 3 by 5, and 4 by 5. How many faces are 3 by 4?

2

Each rectangular face has a congruent opposite face.

Skill: Match congruent faces in pairs

5-6 · Represent Three-Dimensional Figures in Two Dimensions

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

How many faces does a rectangular prism have?
How many matching pairs is that?

A rectangular prism has 6 faces. That forms 3 matching pairs of congruent opposite faces (top/bottom, front/back, left/right).

Compare each part of your work with the worked solution.

6. CHECK B

How many faces does a triangular prism have?
How many are triangles?

5 faces — 2 triangular bases and 3 rectangular sides.

Compare each part of your work with the worked solution.

7. CHECK C

How many faces does a square pyramid have?
How many are triangles?

5 faces — 1 square base and 4 triangles.

Compare each part of your work with the worked solution.

8. TRANSFER

A net has 2 triangles with base 6 and height 4, and 3 rectangles each 6 by 10. What solid is it, and what is its surface area?

A triangular prism. Triangles: $2 \times 12 = 24$. Rectangles: $3 \times 60 = 180$. Surface area = 204 square units.

Explain why your method works and include units when needed.

5-7 · Determine Surface Area of Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Add the face areas $12 + 12 + 15 + 15 + 20 + 20$.

94

Double each pair: $24 + 30 + 40 = 94$.

Skill: Add a list of five or six areas

2. FOUNDATION 2

Find the area of a triangle with base 6 cm and height 9 cm, in square centimeters.

27

Half of 6×9 is 27.

Skill: Find the area of rectangles and triangles quickly

3. FOUNDATION 3

In a rectangular prism with dimensions 2, 3, 5, list the three different face areas.

6, 10, 15

Compute 2×3 , 2×5 , and 3×5 ; each appears twice.

Skill: Read a net and pair up matching faces

4. FOUNDATION 4

One face has area 18 square centimeters. What is the combined area of it and its matching opposite face?

36

Double the face area: 2×18 .

Skill: Double one face's area to cover a matching pair

5-7 · Determine Surface Area of Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

Faces measure 12, 12, 20, 20, 15, 15 square units
— what is the surface area?

Surface Area = $12 + 12 + 20 + 20 + 15 + 15 = 24 + 40 + 30 = 94$ square units.

Compare each part of your work with the worked solution.

6. CHECK B

Faces measure 18, 18, 24, 24, 30, 30 square units
— what is the surface area?

$36 + 48 + 60 = 144$ square units.

Compare each part of your work with the worked solution.

7. CHECK C

A prism has faces 10, 10, 14, 14, 35, 35 square units — what is the surface area?

$20 + 28 + 70 = 118$ square units.

Compare each part of your work with the worked solution.

8. TRANSFER

A cube has surface area 150 cm^2 . What is the length of one edge?

Each of 6 faces is $150 \div 6 = 25 \text{ cm}^2$, so an edge is 5 cm.

Explain why your method works and include units when needed.

5-8 · Determine Surface Area of Pyramids

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Find the area of a triangle with base 8 cm and perpendicular height 7 cm, in square centimeters.

28

Half of $8 \times 7 = 56$ is 28.

Skill: Find the area of a triangle

2. FOUNDATION 2

Find the area of a square base with side 6 cm, in square centimeters.

36

A square has equal length and width: 6×6 .

Skill: Find the area of a square or rectangular base

3. FOUNDATION 3

Four congruent triangular faces each have area 21 square centimeters. Find their combined area.

84

Multiply one face area by four: 21×4 .

Skill: Multiply one triangular face's area by the number of identical faces

4. FOUNDATION 4

A pyramid's vertical height is 9 cm. The altitude drawn within a triangular face is 10 cm. Which height finds that face's area?

10

Use the altitude lying in the triangular face, called the slant height.

Skill: Tell slant height from vertical height on a net or drawing

5-8 · Determine Surface Area of Pyramids

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

One triangular face is 24 square inches. With four identical faces, what is the lateral area?

Lateral area = $4 \times 24 = 96$ square inches.

Compare each part of your work with the worked solution.

6. CHECK B

One triangular face is 30 square inches. With four identical faces, what is the lateral area?

$4 \times 30 = 120$ in².

Compare each part of your work with the worked solution.

7. CHECK C

One triangular face is 18 cm² and the square base is 36 cm². What is the total surface area of the 4-face pyramid?

Lateral: $4 \times 18 = 72$. Plus base: $72 + 36 = 108$ cm².

Compare each part of your work with the worked solution.

8. TRANSFER

A square pyramid has base edge 10 cm and slant height 12 cm. Find the total surface area.

One triangle: $\frac{1}{2} \times 10 \times 12 = 60$. Four triangles: 240. Base: $10 \times 10 = 100$. Total: 340 cm².

Explain why your method works and include units when needed.